

Yashvardhan Khanna

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EDUCATION

SRM Institute of Science and Technology

Chennai, TN

B.Tech in Computer Science (Cloud Computing) — CGPA: 7.97/10.0

2023 – 2027

Delhi Public School

Jodhpur, RJ

Class XII (CBSE): 83.4% — Class X (CBSE): 90.2%

2021 – 2023

TECHNICAL SKILLS

Languages & Core: TypeScript, Java, Python, SQL, JavaScript, C++, OOP, Data Structures & Algorithms

AI-Native & Frameworks: LLMs, Prompt Engineering, Next.js, React.js, Node.js, MERN Stack, REST APIs

DevOps & Cloud: AWS (EC2, Lambda, S3, VPC, Cognito, DynamoDB), Jenkins, Terraform, Docker, GitOps

Tools & Automation: Git/GitHub, Postman, Linux (Bash), Prometheus, Grafana, ArgoCD, n8n SaaS Automation

ENGINEERING PROJECTS

Smart Serverless AI Music App with Context Filtering | *Next.js, SageMaker, Lambda, DynamoDB*

- Owned end-to-end engineering of an AI-native SaaS application integrating web frontends with backend automated data pipelines using AWS Glue and PySpark.
- Deployed an Amazon SageMaker KNN machine learning model to serve personalized, real-time contextual recommendations, delivering a **20% discovery boost** across multi-channel customer test streams.
- Built a low-latency context filter leveraging DynamoDB TTL configurations to alter backend routing configurations automatically under fluctuating user environments with sub-second latency responses.
- Developed a Next.js responsive interface managing background states globally via AudioContext, embedding debounced user input logic and optimistic updates to maximize user interface feedback loops.

Automated AI-Native DevOps CI/CD Pipeline | *AWS EKS, Jenkins, Terraform, Argo Rollouts*

- Designed and implemented a scalable, high-velocity infrastructure using Jenkins and GitOps workflows to automate and accelerate code shipping targets while guaranteeing **100% production application availability**.
- Programmed automated validation stages utilizing custom Analysis Templates, dropping infrastructure rollback cycles to **under 60s** upon spotting live runtime application faults.
- Integrated system metrics directly with Slack notifications using GitOps telemetry hooks, driving cross-functional software sprint resolution speeds forward by **40%**.

IoT Security: Stream Cipher for LoRa E2EE | *ESP32, ChaCha20, Agile Product Development*

- Developed a 256-bit lightweight end-to-end encryption protocol for localized embedded endpoints, successfully protecting internal data transfers without relying on traditional blocking middleware.
- Architected an asynchronous 96-bit Dynamic Nonce synchronization mechanism, saving memory and processing overhead on target devices by eliminating **20% packet bloat**.
- Maintained the entire development lifecycle utilizing agile methodologies, tracking design requirements through sprints to ensure clean code execution and an **18% device energy efficiency boost**.

Corporate Customer Data Platform Warehouse | *SQL, Data Warehousing, ETL Triggers*

- Designed a robust BCNF-normalized data warehouse schema, reducing system data redundancy by **35%** while optimizing query performance across relational tables.
- Automated backend validation tracking utilizing transactional triggers and cursors to harden the safety of operational log lookups and database transaction accuracy.

CERTIFICATIONS & LEADERSHIP

Certifications: Cisco Ethical Hacker; Pursuing AWS Solutions Architect; Python, SQL, and Cloud Architecture.

Leadership: Head Boy (School Leadership); MUN Organizing Team; National-level Athlete.